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**CSA0918-Java Programming**

**CO4-AT1**

### **Question**

A customer-support ticketing tool's window has a toolbar that must always hug the top edge, a status bar that must always hug the bottom edge, and a resizable central content area between them. Identify the single layout manager that naturally expresses this five-region arrangement, and map each screen element to its named region.

### **Aim**

To demonstrate the use of the Border Layout manager in Java Swing by arranging a toolbar at the top, a status bar at the bottom, and a resizable central content area in the center of a customer-support ticketing window.

### **Algorithm**

1. Start the program.
2. Create a JFrame for the customer-support ticketing window.
3. Set the frame layout to BorderLayout.
4. Create a toolbar and add it to the NORTH region.
5. Create a status bar and add it to the SOUTH region.
6. Create the main ticket content area and add it to the CENTER region.
7. Use the WEST and EAST regions for the ticket list and customer details panels to demonstrate all five Border Layout regions.
8. Set the frame size and make the window visible.
9. Stop the program.

## Java Program

```
import javax.swing.*;
import java.awt.*;

public class TicketingLayout {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Customer Support Ticketing Tool");

        frame.setLayout(new BorderLayout());

        JPanel toolbar = new JPanel();
        toolbar.add(new JButton("New Ticket"));
        toolbar.add(new JButton("Save"));
        toolbar.add(new JButton("Close"));
        frame.add(toolbar, BorderLayout.NORTH);

        JList<String> tickets = new JList<>(
            new String[]{"Ticket 101", "Ticket 102", "Ticket 103"});
        frame.add(new JScrollPane(tickets), BorderLayout.WEST);

        /
        JTextArea content = new JTextArea(
            "Select a ticket to view its details.");
        content.setLineWrap(true);
        frame.add(new JScrollPane(content), BorderLayout.CENTER);

        // EAST - Customer Details
        JPanel customer = new JPanel(new BorderLayout());
        customer.add(new JLabel("Customer Details"), BorderLayout.NORTH);
        customer.add(new JTextArea(
            "Name: Customer\nEmail: customer@email.com"),
            BorderLayout.CENTER);
        frame.add(customer, BorderLayout.EAST);

        JLabel status = new JLabel("Status: Ready");
        frame.add(status, BorderLayout.SOUTH);

        frame.setSize(700, 400);
```

```
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

### Sample Input

The user interacts with the GUI by selecting a ticket, viewing its details, and using the toolbar buttons.

### Sample Output

A customer-support window is displayed with the following arrangement:  
NORTH: Toolbar WEST: Ticket List CENTER: Resizable Ticket Content  
EAST: Customer Details SOUTH: Status Bar.

## Explanation

Border Layout is the correct layout manager because it divides a container into **five named regions: NORTH, SOUTH, EAST, WEST, and CENTER.**

- **NORTH** → Toolbar remains at the top.
- **SOUTH** → Status bar remains at the bottom.
- **CENTER** → Main ticket content area uses the remaining space and resizes automatically.
- **WEST** → Ticket list.
- **EAST** → Customer details.

The **CENTER region is especially important** because it expands and contracts when the window is resized.

### Key Point for Assessment

Border Layout is the suitable layout manager because it naturally provides five regions: NORTH, SOUTH, EAST, WEST, and CENTER. The toolbar is placed in NORTH, the status bar in SOUTH, and the resizable central content area in CENTER. The WEST and EAST regions can be used for supporting components such as a ticket list and customer details.

### Result

Thus, the customer-support ticketing interface was successfully arranged using Border Layout. The toolbar stays at the top, the status bar stays at the bottom,

and the central ticket content area automatically expands and contracts with the window. The five Border Layout regions were correctly mapped to the screen elements

## Output Layout







